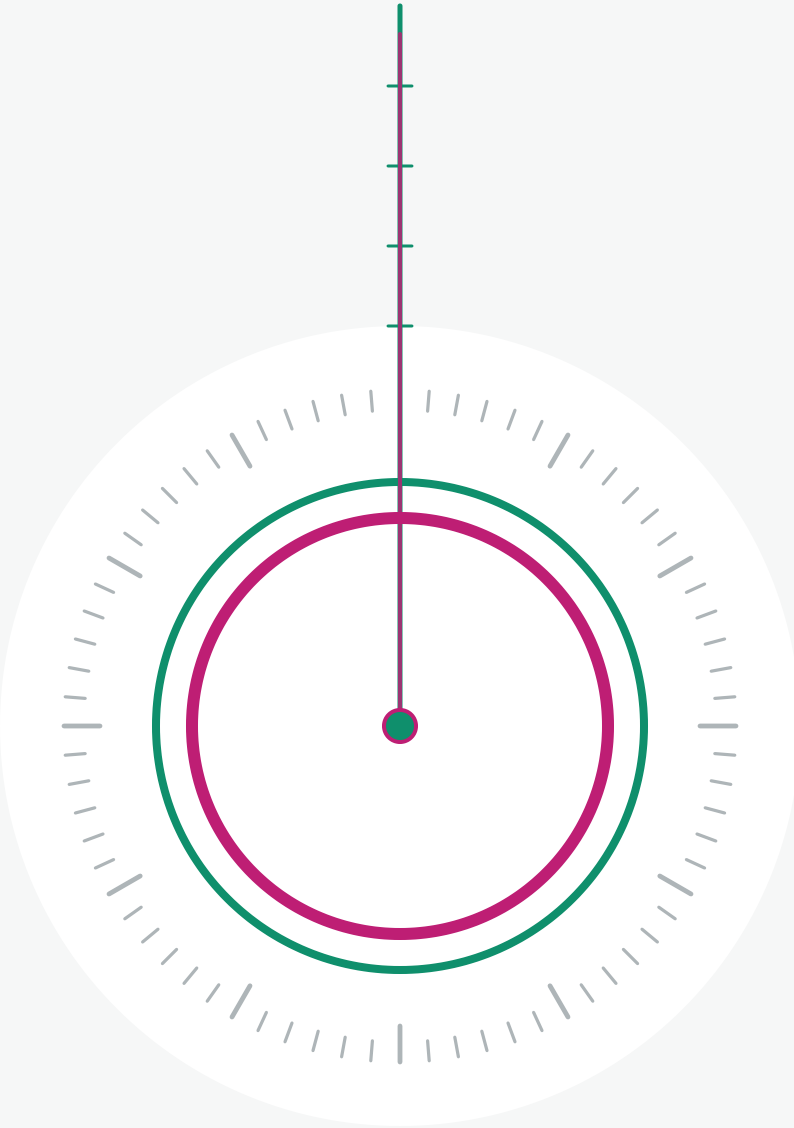
DATUM LINE

Every lab's cells. One "normal".

Every batch, aligned to every lab.



Kat Titterton
Cofounder and CSO, San Francisco



macrofluidic.me/resume

8
peer-reviewed papers
two in Nature Communications

2
patent applications
Axiom Bio, bioactivity detection

100,000+
compounds imaged
two Cell Painting datasets

Kat Titterton

2023–2026 **Axiom**
cell imaging, AI

2022–2023 **insitro**
automated screening

2021–2022 **MIT**
expansion microscopy

2018–2021 **abbvie**
neuron assays

2017–2018 **MGH**
electrophysiology

2017 **COLGATE** BA
neuroscience, math

MY WORK

1 Cell Painting in primary human hepatocytes
Cell Systems 2026, Ewald, Broad Institute

2 Counting cells predicts bioactivity benchmarks
Nat Commun 2026, Seal, Univ. of Cambridge

3 Cell Painting of hepatotoxic botanicals
bioRxiv 2026, Ollitrault, EMBL-EBI

4 An antibody to pathological tau
Mol Neurodegen Adv 2026, Yanamandra, AbbVie

5 Capturing synaptic proteopathic seeds
iScience 2026, Sebastian, AbbVie

6 A neuronal tau aggregation assay
J Biol Chem 2025, Wu, AbbVie

7 Multiplexed imaging of brain nanostructures
Nat Commun 2024, Kang, MIT

8 Tau aggregation in human iPSC neurons
iScience 2022, Manos, AbbVie

9 Neurosteroid photolabeling of $GABA_A$ receptors
Eur J Med Chem 2019, Wu, U Illinois Chicago

Do my cells look “normal”?

asked @ five benches
… answered @ none

“Normal” compared with whom?
Each lab’s own checks can pass while labs still disagree (slides 03 and 04, A13).

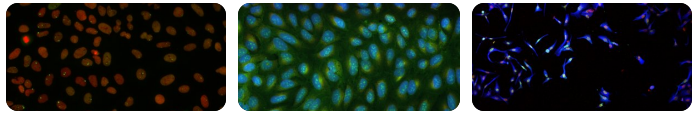
Each lab judges “normal” alone.
Together, every lab can know.

$28B a year of US lab research others cannot reproduce [6]

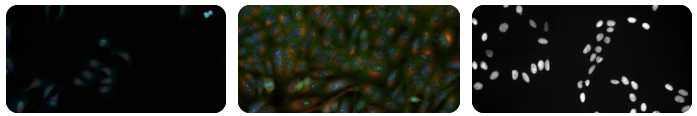
36% ≈ $10B, cells and reagents	64%, other causes

search

U2OS cells “normal” morphology



Doil 2009 | Gustafsdottir 2013 | Stadler 2012



Rohban 2017 | Bray 2017 | JUMP 2023

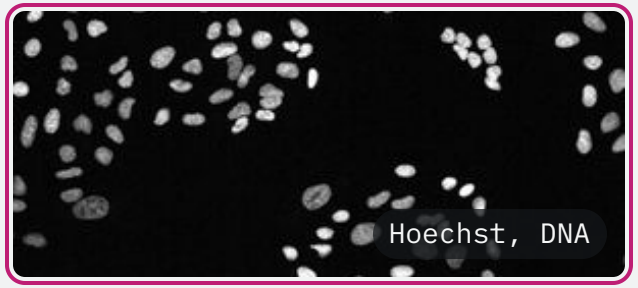
six papers, six “normals”

negative-control wells, IDR idr0010, idr0036, idr0025, idr0033, idr0016 (results/search-tiles.json); JUMP [7]; Doil and Rohban CC BY 4.0, Gustafsdottir and Bray CC0, Stadler CC BY-SA 3.0

ask

To: authors, U2OS paper

Subject: do my U2OS cells look right?



screenshot.png

one opinion

guess

RUN IT

$100,000s

at risk

about $50k a run, 3 runs a week, in the author’s experience [28]

WAIT

months

lost

asking or re-running to be sure, the author’s estimate [28]

One lab, or eleven?

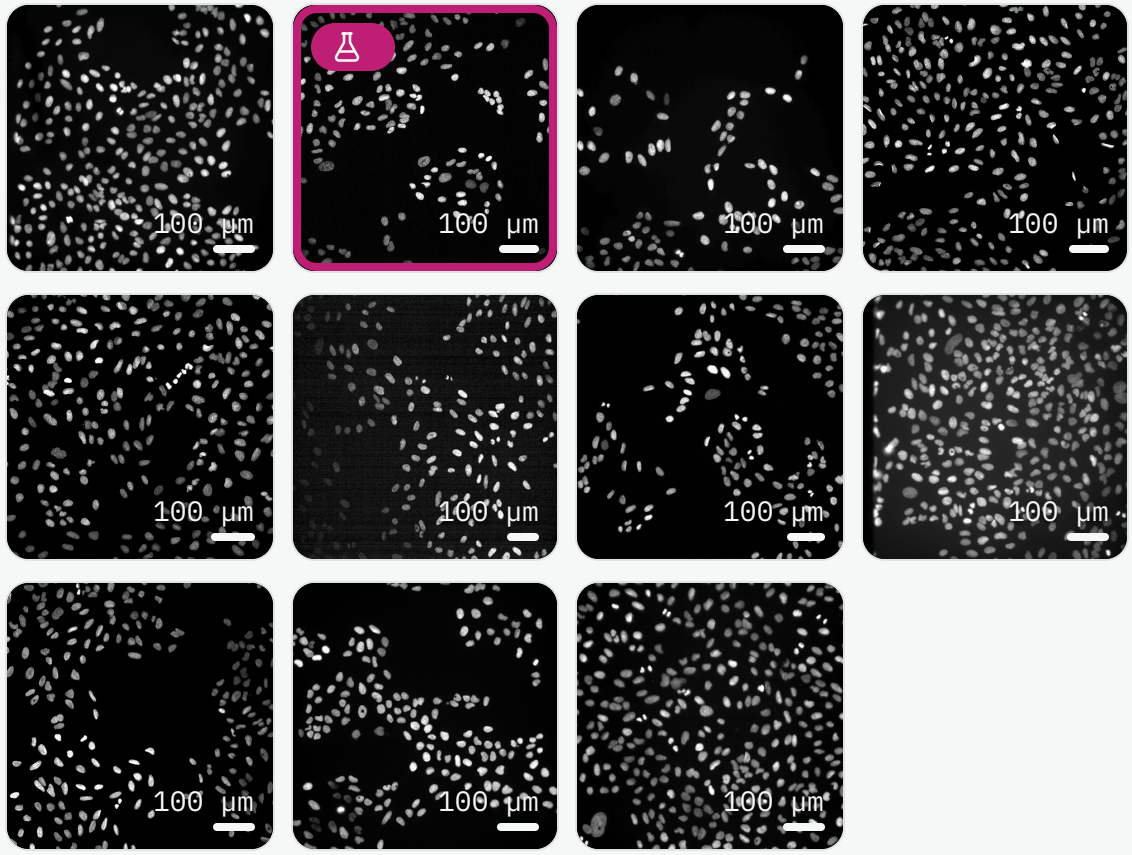


fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

same cells

U2OS: human bone cancer cells

osteosarcoma (bone cancer) from a 15-year-old's tibia, described in 1967 [26]

the one line in JUMP's full dataset, so every lab here grew it [27]

chosen over A549 by a slim margin, for its public data [27]

named in 11 of the Cell Painting Gallery's 28 datasets [17]

a cancer line: "normal" means its DMSO baseline, not healthy tissue

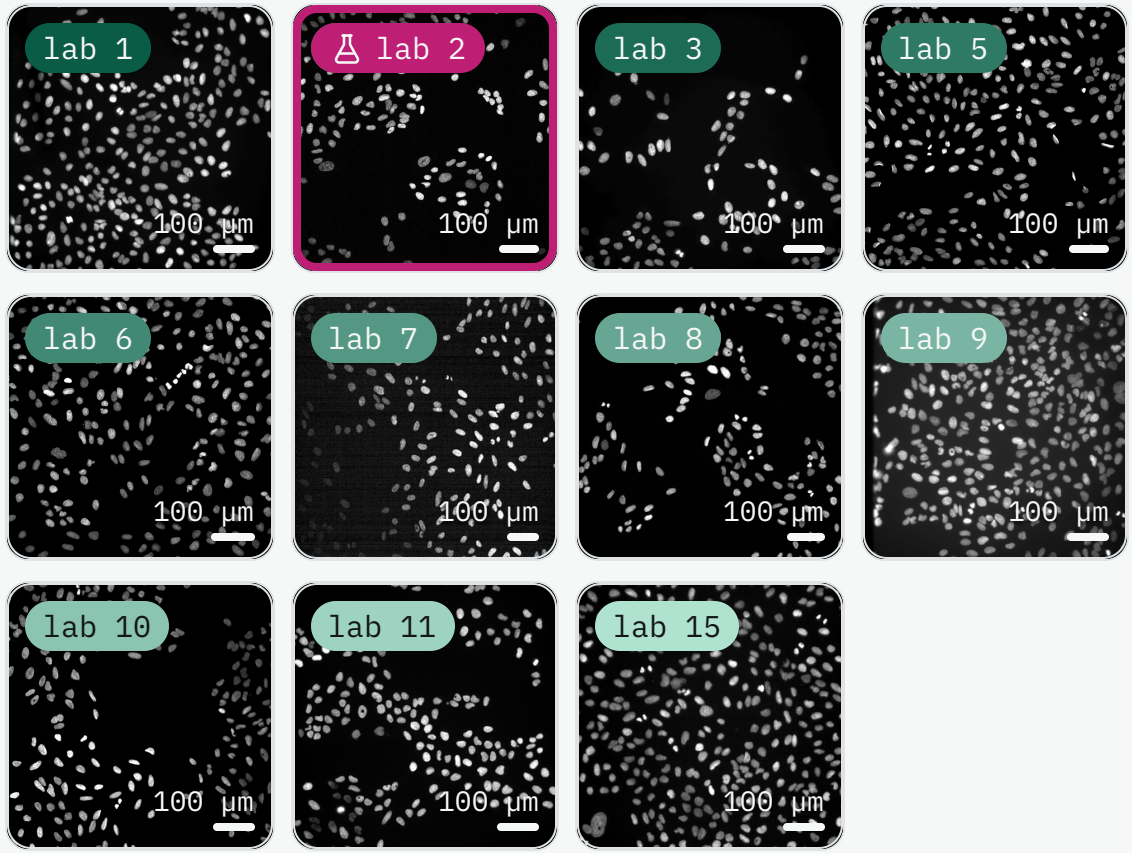
same protocol

Cell Painting, one protocol at every site [7, 27]

no drug

DMSO only, the drugs' solvent [7]

Could anyone tell?
A classifier can.



99.9% right lab, of 11

9% chance

Biosecurity: labels can lie, cells cannot. Good actors can prove what they test. A20

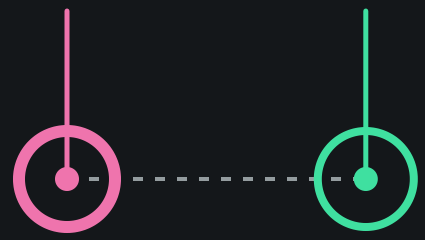
fig 1 Cell Painting, images: DNA channel, classifier: 3,773 numbers per well from all channels, 93,228 wells, 11 labs, plates held out, bars 100 µm, JUMP [7], A2, A3, A5

Eleven labs.
Eleven “normals”.



one well → 3,773 measurements → one point

6× plate spreads apart



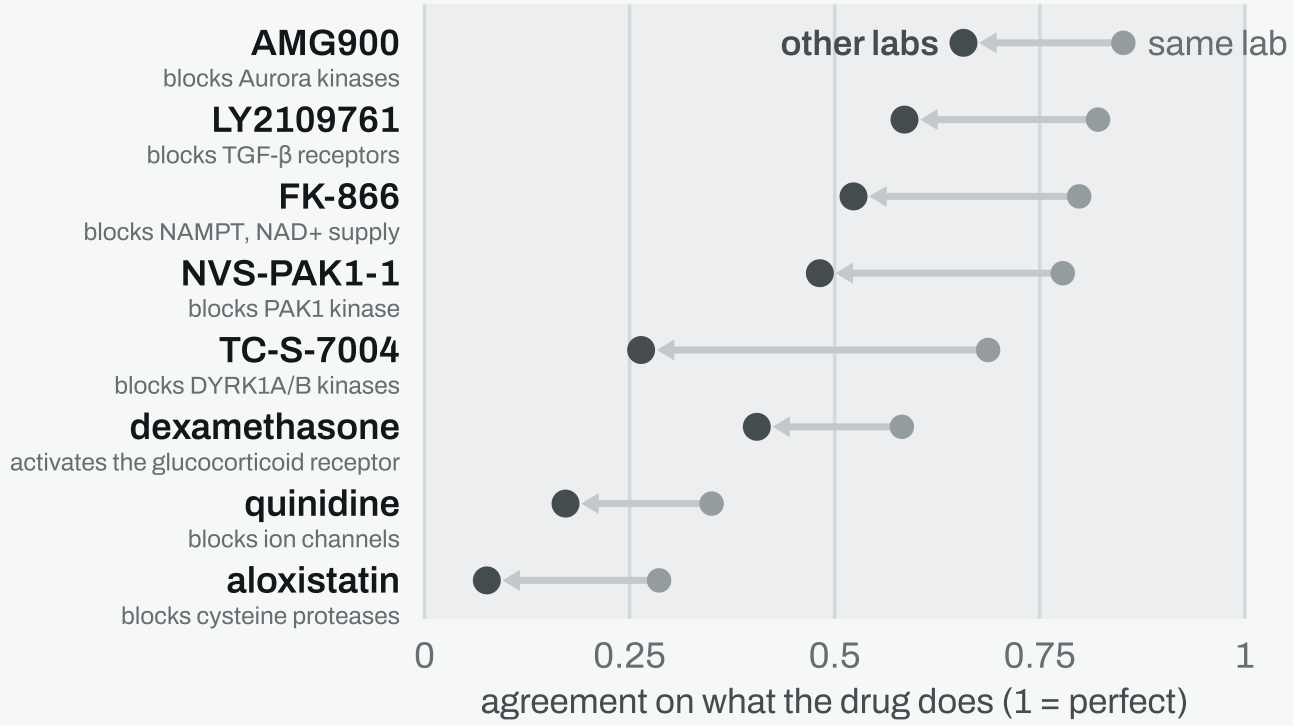
ring: one plate

6.2: the mean distance between two labs’ centres, in units of a well’s mean distance from its own plate’s centre; all 3,773 numbers (results/e13)

fig 2 dots: DMSO wells, 8 per plate, one in three drawn; PCA of 15,352 wells, for the eye (distances use all 3,773 numbers); plates: 384 wells, 1,536 in labs 1 and 9; strip: a lab 2 DMSO well, Hoechst, DNA; JUMP [7]; A2, A3, A13

On one map, labs agree more on what known drugs do.

Eight known drugs, 5 µM in every lab.



mean agreement between plates: same lab 0.64 → other labs 0.40

ON THE MAP

107 of 125

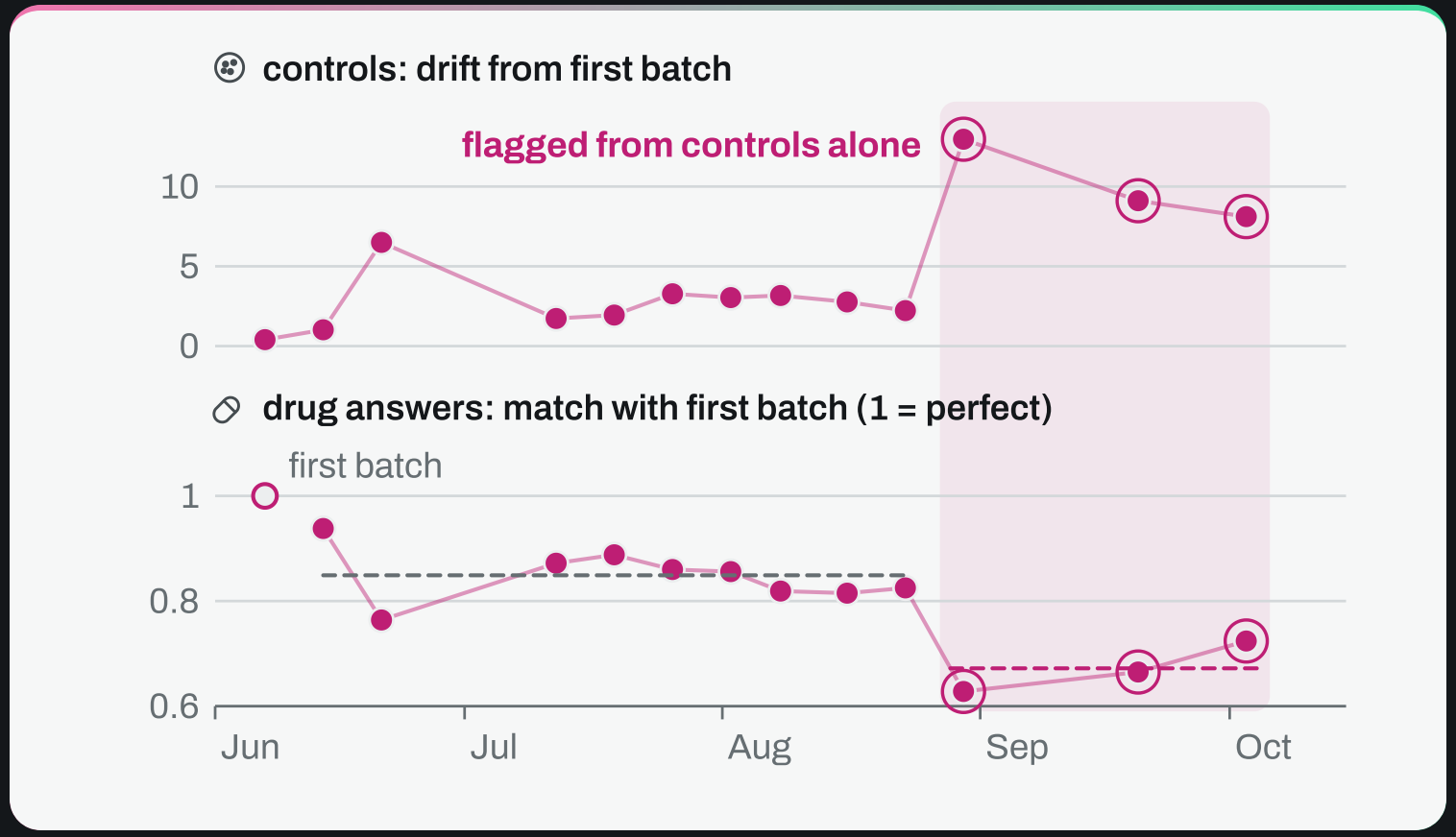
batches agree more with other labs

than on their own controls

by chance: about 62

fig 3 plate pairs, same lab and across labs, 5 µM [7], 1,829 plates, 11 labs (LY2109761 in 10), targets [24]; 7 drugs labs agree on: 0.75 raw, 0.76 usual fix, 0.78 on the map; RSL3 A9; A14, e8, e12

Controls alone catch a drifting batch, before any drug is read.



Lab 2

By autumn it had drifted. From 30 Aug to 3 Oct, 3–5× its summer level (results/e10).

0.85 → 0.67

--- before --- flagged

match with the first batch: nine batches 14 Jun to 23 Aug, three flagged 30 Aug to 3 Oct

across eleven labs: ρ = −0.71, 118 batches, each lab centred (e10, A6)

batch: one run of plates

fig 4 lab 2, 13 batches; drift: control wells' distance from the first batch, in within-plate spreads; match: seven known drugs, 5 µM; dashed: means; flagged: most plates nearer another lab than the first batch (e9); A6; JUMP [7]

The product is the data: every lab's controls, comparable on one map.

labs upload, earn access → **one map** → AI teams pay, by API

SCIENTISTS

Onboard new lines with confidence. Check work against every lab.

$100,000s

a failed screen, caught early

README [28], the author's estimate; drift caught from controls, 04c

AI RESEARCHERS

One dataset from every lab, by API.

24%

lab fingerprint once aligned, from 99.8%

A3, e1 red team; chance 9%

THE FIELD

Results that replicate across labs.

2.2×

sharper "normal" with five labs

06, e13: error 6.2 to 2.8

Proposed terms: labs keep their data and a share of every subscription. A15, A24.

93,228

control wells, every lab

93,196 of them on the map; one plate has no batch (results/e1, e13)

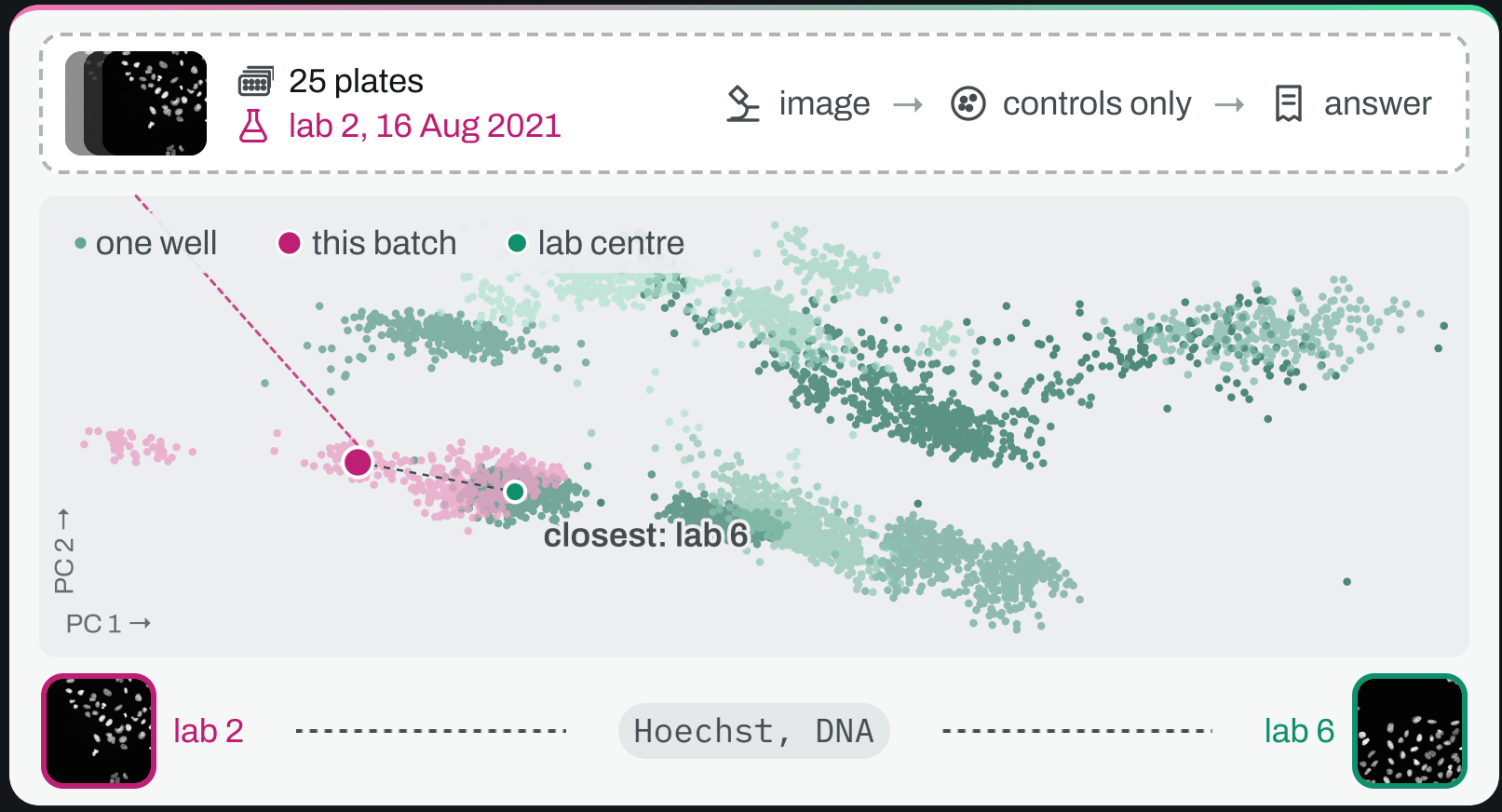
One subscription: the repository, the map, the benchmark.

Public archives keep finished datasets (656 TB, free [17]). This keeps every lab's live controls, aligned and checked.

Every cell result becomes checkable against the field.

the product as proposed, counts from results/e1 and e13, JUMP [7], A15, A18, A24

Image the plate, send the controls.
Know it’s “normal” before the screen.



Certificate

Do my controls look “normal”?

Do my drugs match other labs?

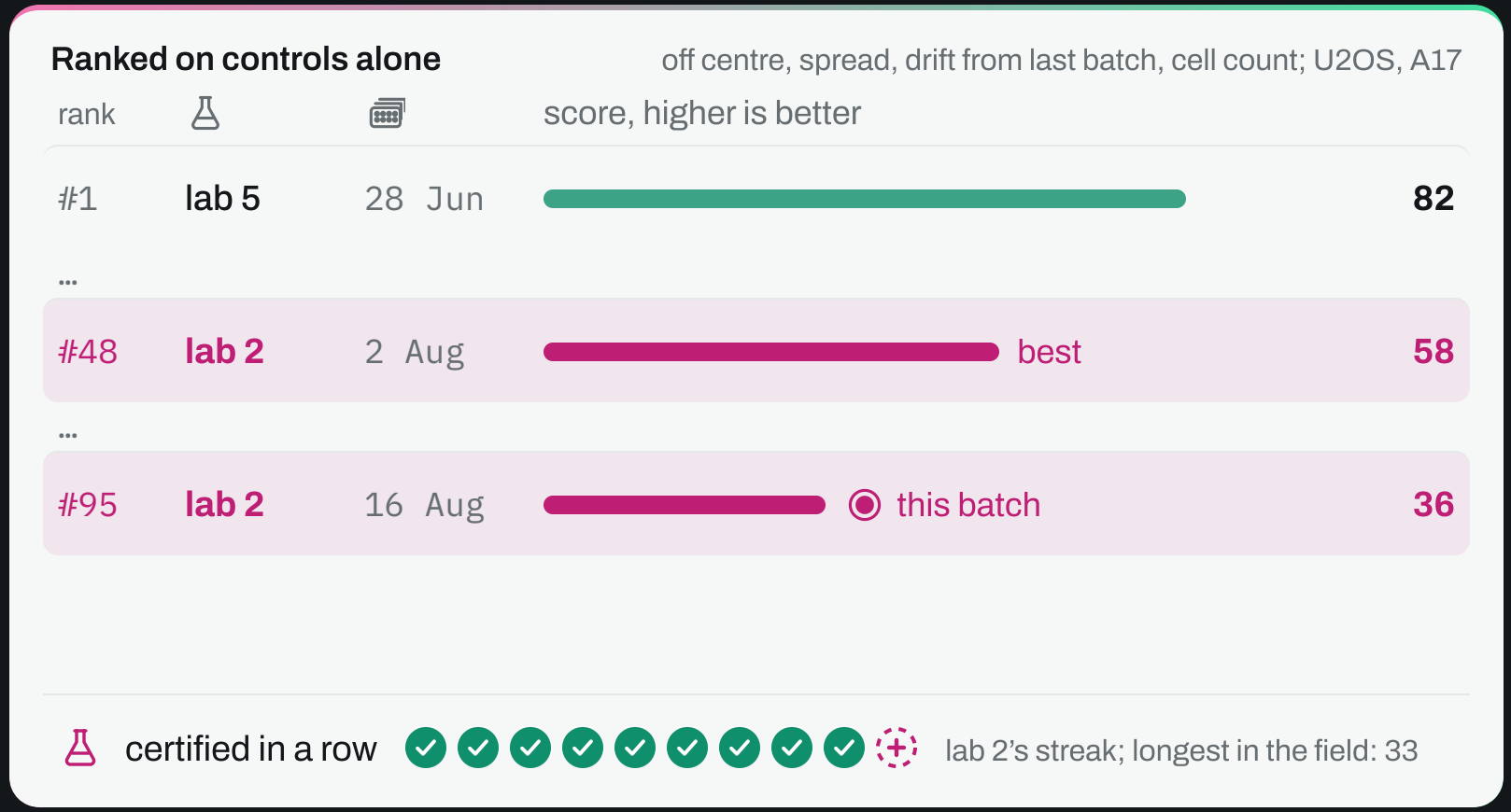
known drugs worked; agreement with other labs: 0.70 measured, **0.77** on the map (1 = same)

Did my cells grow as usual?

in range: 188 cells per well

fig 5 a proposed certificate, replayed on public data; closest lab in all 3,773 numbers: lab 6, 4.4 within-plate spreads away; JUMP [7], results/e9, e11, e12; A11, A18

Which labs make the most reliable data?
Ranked on controls alone.



Standing

#95 of 129 batches

Batches certified

10 of 13

To climb: steadier batches

Lab 2's drift from last batch is more than twice the field's.

2.9

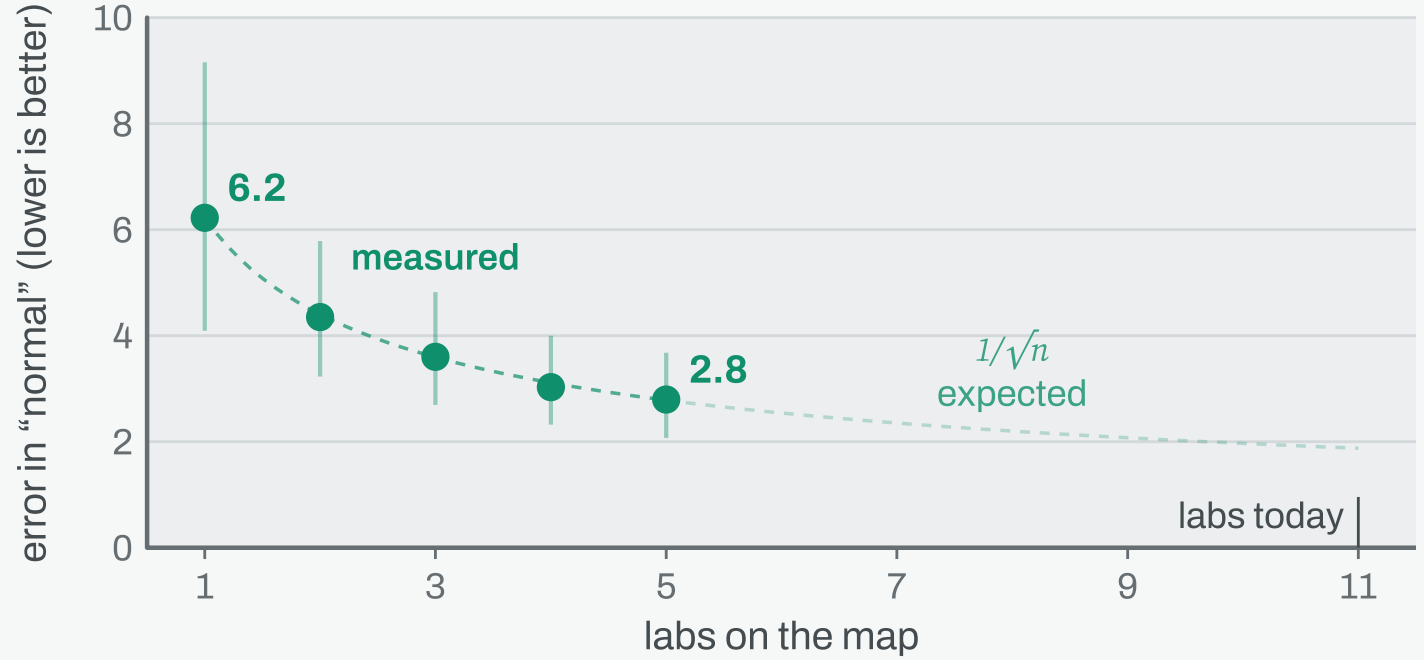
1.2

drift from last batch, lower is better: lab 2, then the field's median batch

private, unless the lab shares it

fig 6 score = 100 − mean percentile of four measures, DMSO wells, every batch: 129 (05b counts 125: it leaves out the four lab 3 batches whose known drugs failed, e11, e12); JUMP [7]; A17

Every lab that joins sharpens “normal” for everyone.



Map, today

↓

Certificate, next

↓

Subscriptions, then

↓

Journals, raw images, every paper

↓

…

fig 7 error: distance between two maps from different labs, in within-plate spreads; dot: mean of 400 draws, bar: 10th to 90th percentile; measured to five labs, dashed beyond; 1,871 plates; JUMP [7]; results/e13

Every institution adds revenue. 100 of them: $15M a year.

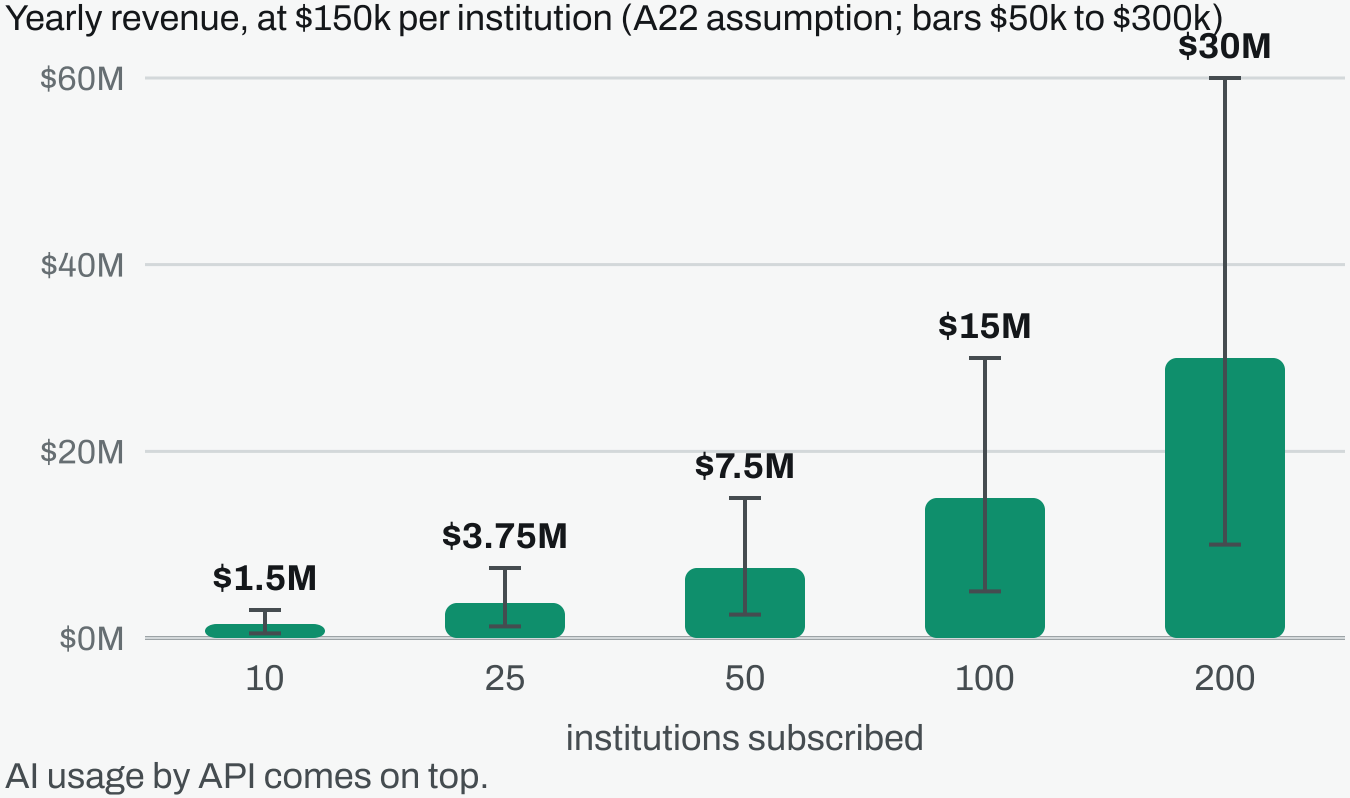


fig 8 revenue = institutions × site licence, $150k a year assumed (A22); bars: $50k to $300k; AI usage by API not priced; EU-OPENSCREEN on A23

$30M

one cell dataset

Roche paid it: a milestone for one whole-genome map of microglia [11]

Why now

- 2023 — 1.8 billion cells, public [7, 25]
- 2025 — FDA moves off animal tests [9]
- now — AI reads cell images, A10

JUMP went public from 2023 [7] and counts 1,810,840,525 cells today; source_15 is not counted, so these are not this map's eleven labs [25]

Six questions, six tools.
Use all of them.

Each answers one question well.
Each misses something, Datum Line too.

tool	answers	misses
DNA fingerprint (STR)	Is it the line the label says?	how the cells behave A7
vendor certificate	Was it fine when it shipped?	anything after shipping
plate controls	Did this plate work?	a whole lab drifting A1
in-house platforms	Is my own data consistent?	lab differences; it removes them A3
data factories	Where do I get cell data at scale?	other labs' "normal" A23
Datum Line	**Do my cells look "normal", next to every other lab?**	which cell line it is; unstained images, not yet A7, A10

STR: ATCC [14]

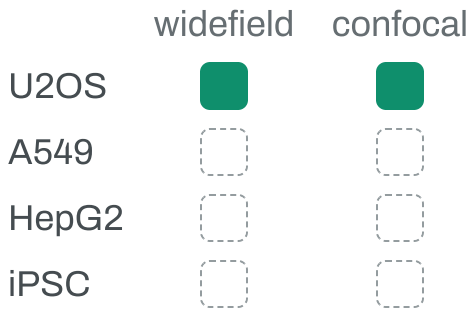
WHAT THE MONEY BUYS

$1.8M

pre-seed, $0.9M line by line, ×2 for the usual overrun, A21

Eighteen months to first revenue.

1 → 2 → 3 → 4 → seed round



benchmark playground

known drugs, planted faults, A4, A16

11 + 3

labs today + 3 partners

three partner labs on the map

U2OS first, then any line



99.5%

right lab of 8 labs with brightfield

model for any image

from one brightfield image; chance 12.5%, A10, A19

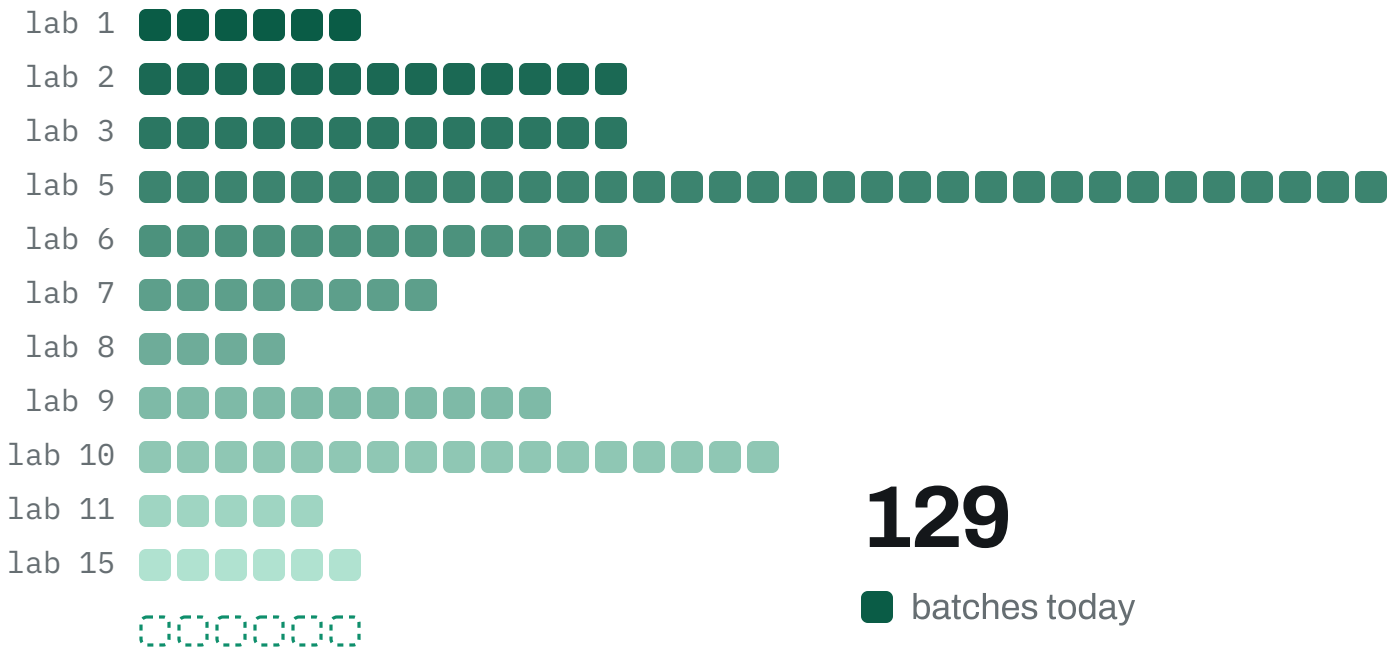
first paid certificate

a company pays ~$10–50k per check, A15

proposed, milestones in order, not dated

Every plate imaged, placed on the map.

Labs share control images, free.
Each sees its “normal” next to everyone else’s.



$1.8M

pre-seed, eighteen months, A21

1 **leader**, to run it

3 **partner labs**, bench time

2 **benchmark partners**

**Have control images?
Talk to me.**

fig 10 one square: one batch, a median of 10 plates (1 to 58); 1,871 plates, JUMP [7], results/e9; A15, A20, A22, A24

The rest, for questions.

Every number comes from the code here or a cited source.
Unfinished work is marked pending. All 25 boards: archive.html

A3 **Why not batch-correct it away?**
erases most of the lab fingerprint

A7 **Isn’t this cell-line DNA testing (STR)?**
which cells, how they behave

A9 **Does it change the result?**
same drug, different answer

A10 **Does it work without stains?**
bet on brightfield

A14 **What does the map buy?**
same drug, closer between labs

A15 **Who pays?**
labs join free, subscribers pay

A18 **What is private?**
control wells leave, nothing else does

A19 **What is the machine learning?**
image in, position and correction out

A20 **What about biosecurity?**
cells show their lab, labels can lie

A21 **How much, and what for?**
planned $0.9M × 2 = $1.8M asked

A22 **How big?**
count × price, three streams

A24 **Why would a lab share?**
owned by the lab, verified by anyone

The usual correction erases most of the lab's fingerprint.

What counts as "normal" depends on every variable below.
JUMP records the left column; nobody records the middle.

RECORDED IN JUMP

- lab
- batch date
- plate
- plate type
- well position
- microscope
- confocal or widefield
- laser or LED
- objective NA
- fields per well
- filter set
- pixel size
- brightfield planes
- CellProfiler version

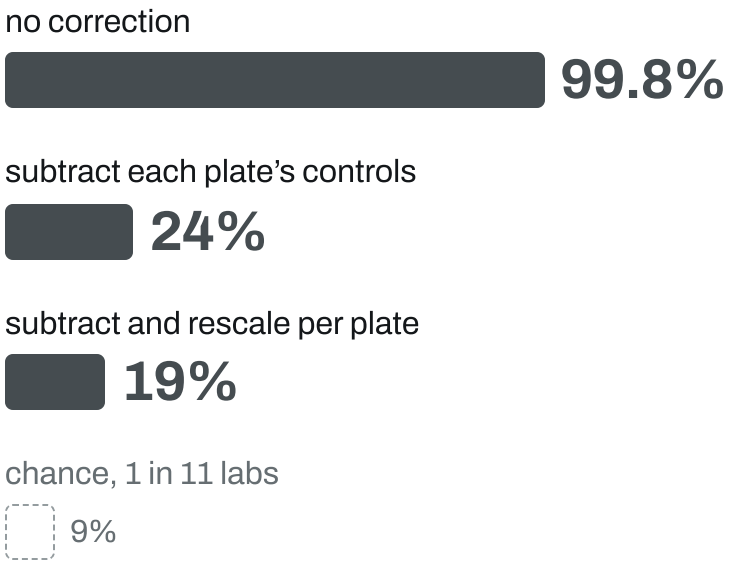
in every image file too:
image size, stored bit depth

NOT IN ITS PUBLIC METADATA

- DMSO % per well
- DMSO lot
- DMSO water uptake
- stock freeze–thaw
- serum lot
- media lot
- passage
- thaw date
- seeding density
- confluence at dosing
- dosing to fixation
- stain and dye lots
- operator
- incubator shelf
- edge evaporation
- mycoplasma status
- CO2 and humidity
- compound dose

not required to publish today
in our own data: tracked, every one

CAN A MODEL TELL WHICH LAB?
HIGHER = CLEARER LAB FINGERPRINT



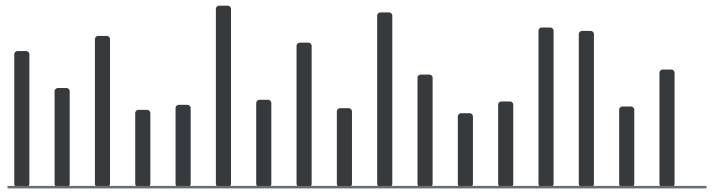
29,943 DMSO wells, 16 per plate
tested on plates it never saw

Batch correction is built to remove this signal [18], variables: JUMP metadata [7]

STR says which line; images say where it was made.

One cell line, U2OS, in all 11 labs.
Images still name the lab: 99.9% right.

STR, DNA FINGERPRINT

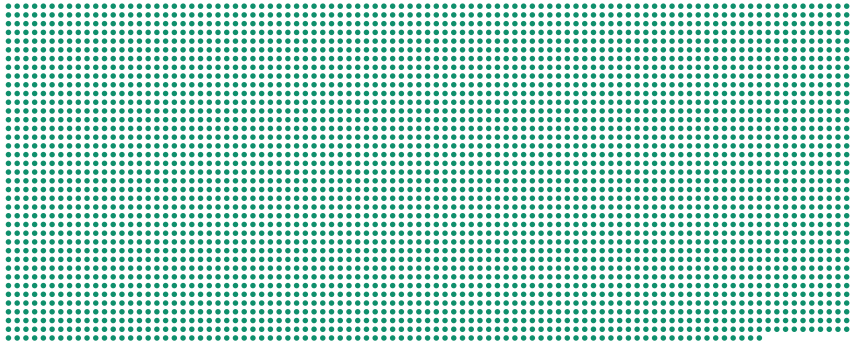


17
DNA markers

periodic
per cell line [14]

Standard test for identity.
Cannot say where or how the cells were grown.

DATUM LINE



3,773
numbers per well

every batch
images already taken

Shows where cells were made, next to other labs.
Cannot say which cell line it is.

identity, STR + behaviour, Datum Line = **one certificate**

natural first partner: STR testing services

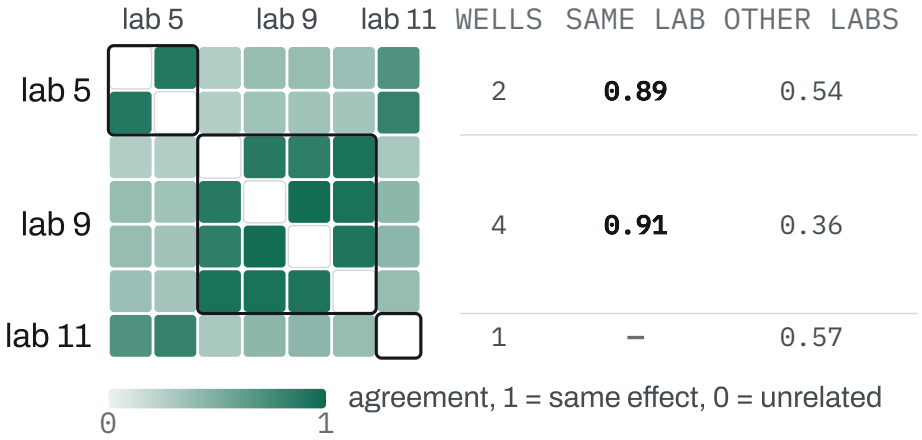
STR: 17 loci + amelogenin, ATCC [14], Datum Line: 93,228 U2OS DMSO wells, 1,872 plates, 11 labs, plates held out, JUMP [7]

The same drug looks less alike in another lab.

RSL3 at 10 µM in three labs, every well against every other.
Darker = more alike.

WHAT A DRUG SCREEN REPORTS — drug wells – DMSO wells, same plate

RSL3 AT 10 µM, EVERY WELL AGAINST EVERY OTHER



	WELLS	SAME LAB	OTHER LABS
lab 5	2	**0.89**	0.54
lab 9	4	**0.91**	0.36
lab 11	1	–	0.57

SAME 7 PLATES, EACH WELL AGAINST ITS OWN PLATE'S DMSO

	SAME LAB	OTHER LABS
RSL3, 10 µM	**0.90**	0.49
8 known drugs, 5 µM, median	**0.74**	0.26
DMSO vs DMSO	**0.00**	0.00

DOES IT CHANGE THE NUMBER?

Which lab ran it explains 20% to 71% of how strong a drug's effect looks.

HOW SOLID?

Seven wells: lab 5 two, lab 9 four, lab 11 one. Lab 11 has no same-lab pair.
Most of the gap sits beside lab 9, on 1536-well plates: 0.33–0.39 against 0.75.
Its known drugs split the same way.

LAB 7, LEFT OUT

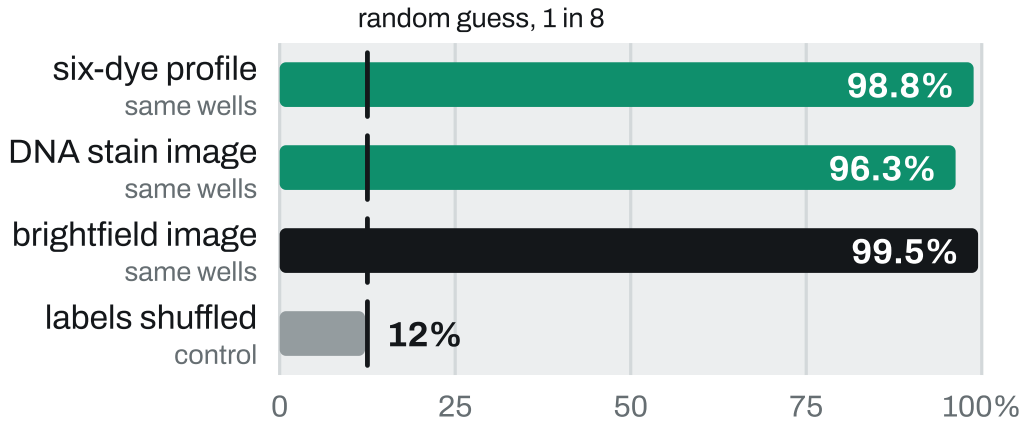
Dosed at 0.625 µM, a sixteenth [7]: RSL3 left 75–87% of its cells.
AMG900 at 5 µM left 36% there, 28–30% elsewhere: its cells respond.

agreement = cosine of two wells' effects, each well minus its own plate's DMSO wells, 8 drugs at full scale: 1,829 plates, 11 labs, JUMP [7]

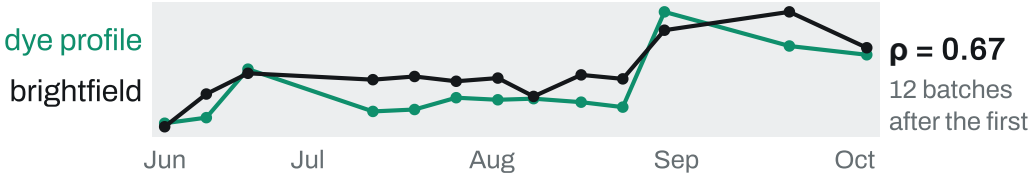
One brightfield image already names the lab.

Evidence: cells stained with six dyes.
Product: plain light, no dye.

CAN ONE WELL TELL WHICH OF 8 LABS IT CAME FROM?



DOES BRIGHTFIELD SEE LAB 2 DRIFT?, EACH LINE SCALED TO ITS PEAK

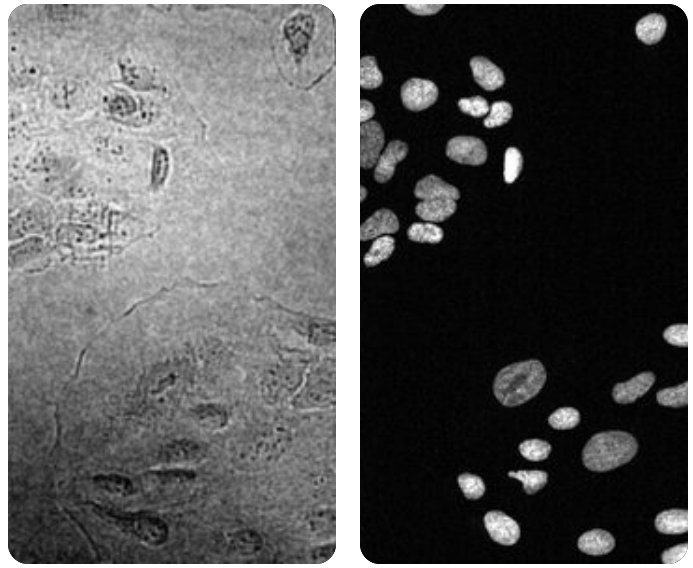


Not yet the whole map: beyond each lab's usual neighbour, brightfield and dye profile agree batch by batch no more than luck (45 of 102 against 41).

WHY THE BET IS FAIR

No dye: fewer things can go wrong

~~stain lot~~ ~~dye lot~~ ~~staining step~~ A3



BRIGHTFIELD DNA STAIN

same spot in one well, lab 2

% named right, each lab counted equally, unseen batches, one DMSO well per plate, 1,338 plates, the same wells and model for every row, median of 20 splits of the batches, profile without its 8 frame-size columns, shuffled: 5 seeds per input, image model never trained on cells, JUMP [7]

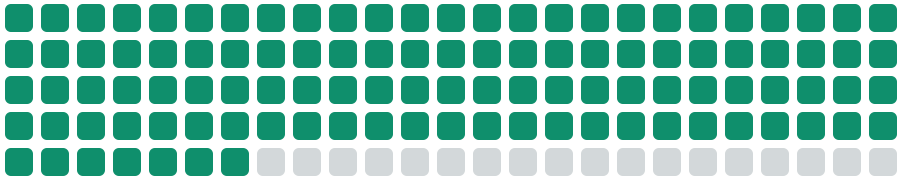
On the map, labs agree a little more on the same drug.

Seven drugs with a known effect, 125 batches, 11 labs.
Higher is better.

Do labs agree on what a drug does?

0.75 raw → **0.76** usual fix → **0.78** on the map

1 = every lab sees the same effect

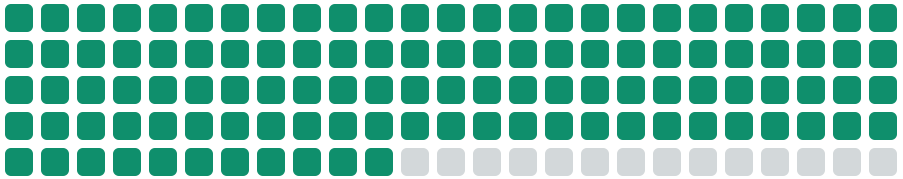


107 of 125 batches higher on the map than with the usual fix
than raw: 121 of 125

Can you still tell the drugs apart?

0.52 raw → **0.56** usual fix → **0.59** on the map

gap to the nearest wrong drug, higher is easier



111 of 125 batches higher on the map than with the usual fix
than raw: 125 of 125

NOTHING LOST

Right drug recognised: 0.92 → 0.94
Matches its own past: 0.88 → 0.88

HOLDS

Labs agree more: 5 of 5 ways of checking
Drugs stay apart: 4 of 5

NOT FIXED YET, DRUG STRENGTH

Varies batch to batch, lower is better
raw 6% → usual fix 20% → map 12%: steadier than the usual fix, not yet as steady as raw

usual fix: each batch set against its own “normal”, its DMSO wells, map: against every other lab’s, 1,829 plates, JUMP [7]

Labs earn access by uploading; AI teams pay the most, by API.

The certificate pays first.
The API is the main revenue: access, not ownership.

WHO	WHAT THEY GET	WHAT THEY PAY	WHAT IT BUILDS
labs and core facilities	access to the repository and the map; certificate, leaderboard	**nothing: uploading controls earns their access**	the map
AI labs and model builders	certified raw images from every lab, by API	**the most: priced by use, not yet set**	main revenue
companies that buy cell data	a certificate on every dataset they buy	**~$10–50k per certificate**	first revenue
teams that run cell screens all year	a check on every batch they run	**~$5–15k per cell line, per quarter**	repeat revenue
benchmark partners	a test set with faults planted on purpose	**~$50–200k a year**	the standard

Only control wells leave the lab.

No compound, no target, no result.
Nothing for legal to approve.

WHAT LEAVES THE BUILDING

control wells only: DMSO and known drugs



green: 64 of 384 wells leave, JUMP [7]

images, or numbers from them

proposed: images deleted once measured

a plate map and the batch date

which wells, and when they were imaged

WHAT DATUM LINE DOES WITH IT

builds the map

one average point per plate.
Scale set by the other labs, never by yours.

the map holds DMSO wells only

82,319

for lab 2: DMSO wells of 10 other labs
0 treated

answers

where you sit, how far you moved, nearest lab, cell count, “normal” or not

ranks every batch

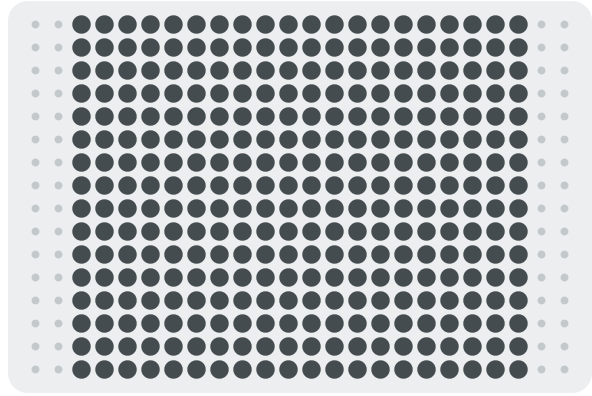
on its DMSO wells, never on results, A17

sends a correction, not a model of you

one per batch, fitted on your DMSO wells, A19

WHAT IT NEVER SEES

your treated wells



dark: 320 of 384 wells stay with you

your results

corrected in your building, by you

your name, unless you choose

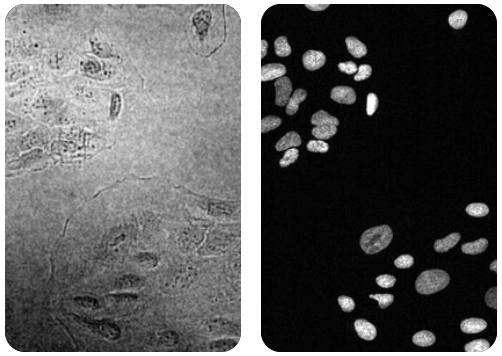
proposed: a number until you opt in.
Every lab here is a number.

In every analysis here, the map and the correction see DMSO wells only, github.com/kat-titter/datum-line/analysis, JUMP [7]

An image goes in; a position and a correction come out.

Results stay in the lab.
DMSO wells and the correction travel.

1, AN IMAGE



brightfield DNA stain

one DMSO well, JUMP [7]

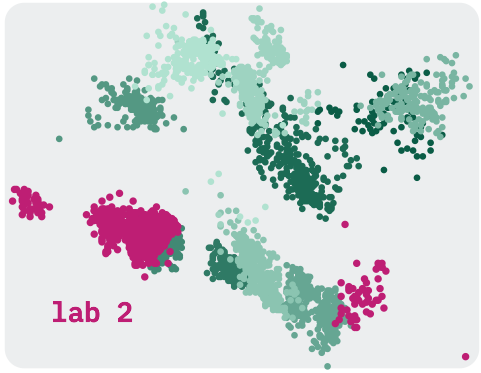
2, NUMBERS

3,773

numbers per well

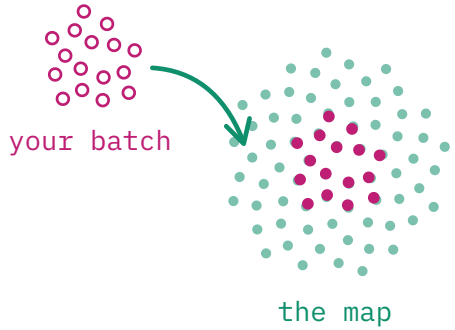
standard software today: size, shape, brightness, texture

3, THE MAP



each lab placed on the other labs' scale, never its own

4, THE CORRECTION



one per batch, fitted on your DMSO wells, sketch, A14

EVERY LAB LOOKS DIFFERENT

99.9%

a computer names the lab of each well, by guessing: 9%
93,228 DMSO wells, 11 labs, on plates it never saw

AI IMAGE MODEL NEXT, A10



right lab, from one picture

Correction, per batch on 200 components: $x' = C_{field}^{½}\, C_{batch}^{-½}\, (x - m_{batch})$, every script: github.com/kat-titter/datum-line/analysis

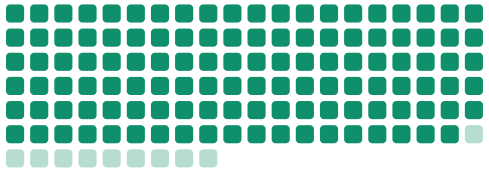
Every lab leaves a fingerprint, so a wrong label stands out.

One map of DMSO wells, no results stored.
Two checks work today, three are proposed.

PROVENANCE today

99.9%

of 93,228 DMSO wells get the right lab, on unseen plates: cells and microscope together.



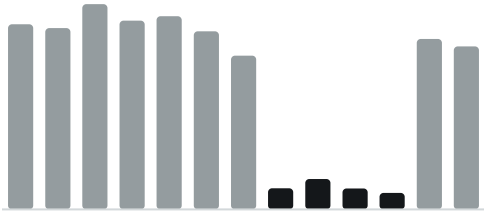
1 square = 1 batch
solid: every well right, 119

INTEGRITY today

4 batches

Label: known drug in these wells.

Cells: almost no effect, on 65 plates.



lab 3, drug effect per batch
dark: failed, A8

UNKNOWN ORIGIN proposed

9×

Farthest of 129 real batches from its nearest other lab.

Farther out: wrong cell line? Mixed culture? Untested.



distance to nearest other lab
1 line = 1 real batch

EARLY WARNING proposed

30 Aug

2021: lab 2's DMSO wells moved; its cell count did not.

A move nobody explains: check reagents, air, water.



lab 2, since batch 1
filled: flagged, A13

DECLARED VS OBSERVED

proposed, opt-in

A company that wants the good-actor mark sends treated wells and says what is in them. The map compares them with JUMP's public profiles of 100,000+ compounds [7] and flags cells that look like a known toxic class. Only what a lab sends is seen.

Integrity check uses each plate's known-drug wells (A8), × = times the usual spread of wells within one plate, JUMP [7]

proposed, every line an assumption

About $1.8M for eighteen months, mostly people.

$884k planned × 2 = $1.77M asked



people 71% | benchmark 17% | other 12%

leader, to run it	18 months	**$180k**
scientific lead (me)	18 months	**$180k**
ML engineer	from month 7	**$160k**
payroll overhead	20% of salaries	**$104k**
benchmark playground	~60 plates, planted faults, A16	**$120k**
partner-lab bench time	3 labs, controls they already image, reagent top-ups, shipping	**$30k**
compute and storage	images, models, hosting, API	**$45k**
legal, IP, insurance, accounting	incorporation to first contract	**$45k**
travel and partner visits	2 benchmark partners, 3 labs	**$20k**
planned	sum	**$884k**
+ buffer	most things take 2–3× the time and money planned, we plan for 2×	$884k
asked, 18 months		**$1.77M**

NO IN-HOUSE LAB

Partner labs or a contract lab.
Own lab: costs more, proves less.

WHAT THE MONEY MUST SHOW

1 benchmark playground, paid in this round **2** three partner labs on the map
3 model for any image **4** first paid certificate → seed

salaries below market, leader and scientific lead paid, nothing here committed

About $84M a year, if every likely buyer joined.

Count × price.
Three streams.

assumption	low	base	high
organisations testing drugs on a few cell lines	200	400	800
cell lines each one uses per quarter	2	4	8
price per cell line per quarter, $k	5	10	15
organisations testing drugs on hundreds to thousands of cell lines	10	25	50
site licence per year, $k	50	150	300
per-line checks and site licences, per year	**$8.5M**	**$67.8M**	**$399M**
companies buying cell data they did not make	50	150	300
purchased datasets each certifies per year	2	4	8
price per certificate, $k	10	25	50
certificates on purchased data, per year	**$1M**	**$15M**	**$120M**
benchmark partners	4	10	20
membership per partner per year, $k	50	100	200
benchmark membership, per year	**$200k**	**$1M**	**$4M**
total per year, if every buyer joined	**$9.7M**	**$83.8M**	**$523M**

BASE CASE, PER YEAR



↑ five-year target, a tenth, $8.4M

benchmark $1M ↑

WHY BUYERS PAY

One screen I ran on a line checked late:
3 runs a week × 17 weeks × $50k ≈ $2.6M to redo.
Certificate @ $25k: about 1% of that.
Subscription: insurance against it.

prices proposed on A15, academic labs join free, they are the map, not the revenue

The lab keeps its data and a share; anyone can check it.

Subscribers get access, not ownership.
No crypto token.

CONTROL WELLS ONLY — today

no test compound, no target, no result
results never leave the lab

LAB KEEPS ITS DATA — proposed

sharing is opt-in
withdraw any time

LOW RISK BY DESIGN — today

cell-line images only
no human genetic data, no pathogen data

LABS LEAVE FINGERPRINTS — today

99.9% right lab out of 11
from control-well images A20

SIGNED AND TIMESTAMPED — proposed

public record, anyone can check
dates cannot be changed, blockchain optional

KNOWN SUBSCRIBERS — proposed

verified at sign-up, odd access flagged
access log open to members

lab shares
images of control cells
certificate attached

→

certificate checks
how “normal” the cells look
which lab, what changed

→

subscriber pays
to use certified data
from many labs

→

lab keeps a share
of every subscription
as long as its data is used